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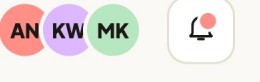
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Adam
Nowak

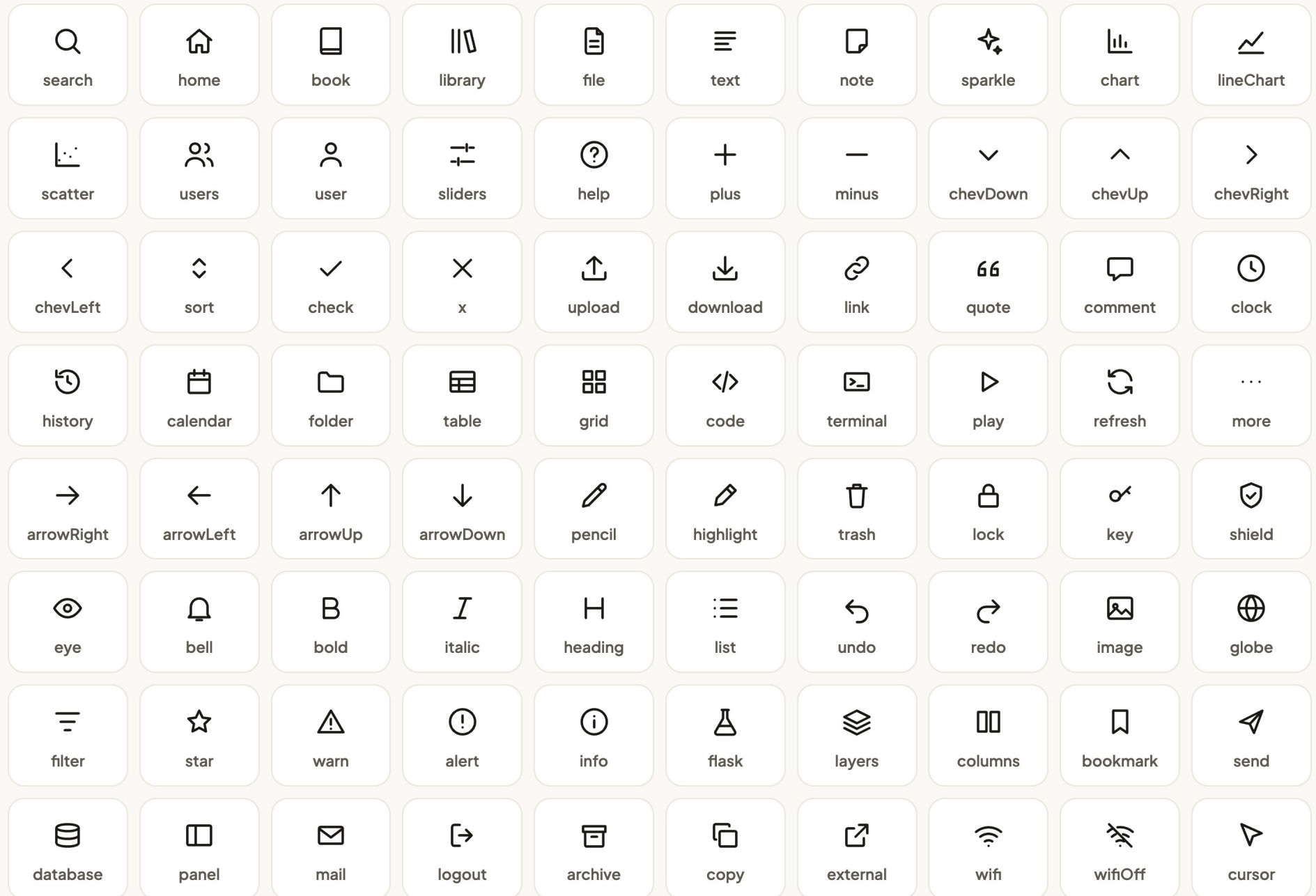
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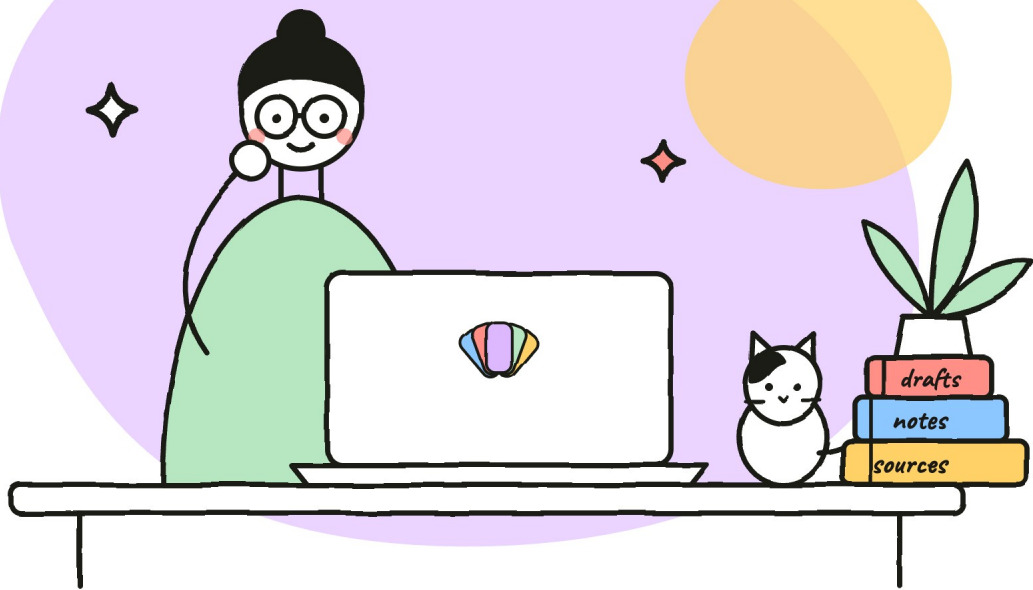


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24 px grid · 1.75 px rounded stroke · inherits currentColor. Functional icons stay systematic; hand-drawn art lives in illustrations only.



Ask. Check. Cite.



- Laboratory Report
- Research Notes
- Draft Conclusions

OUTLINE

- Abstract
1. Objective
2. Theoretical Background
3. Methodology
4. Results
5. Analysis
6. Conclusions

CONTENT ORIGIN

Human written

AI generated / rewritten

Source-derived claim

Analysis-derived claim

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Effect of Temperature on Photovoltaic Cell Efficiency

Laboratory Report · Adam Nowak, Kasia Wiśniewska, Michał Kowalczyk · v14

Abstract

This report examines how cell temperature affects the conversion efficiency of a monocrystalline silicon cell. Measurements were taken between 25 °C and 65 °C under constant illumination of 1000 W/m².

1. Objective

Determine the temperature coefficient of efficiency for the test cell and compare it with values reported in the literature.

2. Theoretical Background

Photovoltaic efficiency decreases as temperature increases. ¹ The open-circuit voltage falls by roughly 2.2 mV per kelvin, while the short-circuit current rises only slightly ². As a result, the fill factor and the maximum power point both shift downward.

- Voc decreases by about 2.2 mV/K (lecture_05.pdf, p. 22)
- Isc increases by about 0.05 %/K
- Net efficiency loss of about 0.41 %/K

3. Methodology

We put the cell on the plate and made it hotter step by step and wrote down the numbers.

4. Results

Table 1 — Electrical parameters of the test cell at 1000 W/m²

T (°C)	Voc (V)	Isc (mA)	η (%)
25	0.612	34.8	19.4
35	0.589	35.1	18.6
45	0.566	35.4	17.8
55	0.543	35.8	17.0
65	0.520	36.1	16.2

10 kΩ